

Perimeter and Area of Figures

1.

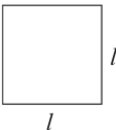
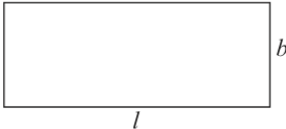
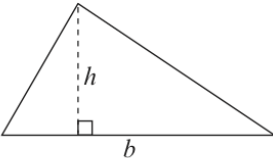
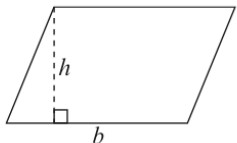
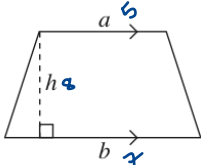
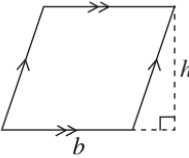
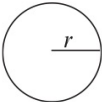
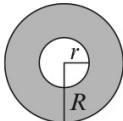
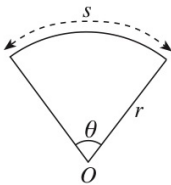
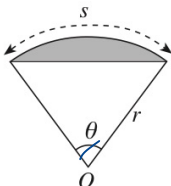
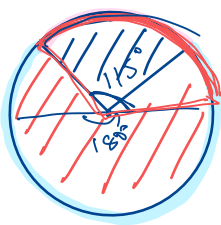
Figure	Diagram	Formulae
Square		Area = l^2 Perimeter = $4l$
Rectangle		Area = $l \times b$ Perimeter = $2(l + b)$
Triangle		Area = $\frac{1}{2} \times \text{base} \times \text{height}$ $= \frac{1}{2} \times b \times h$
Parallelogram		Area = base $\times$ height $= b \times h$
Trapezium		Area = $\frac{1}{2} (a + b) h$
Rhombus		Area = $b \times h$

Figure	Diagram	Formulae
Circle		Area = πr^2 Circumference = $2\pi r$
Annulus		Area = $\pi(R^2 - r^2)$
Sector		Arc length $s = \frac{\theta^\circ}{360^\circ} \times 2\pi r$ (where θ is in degrees) $= r\theta$ (where θ is in radians) Area = $\frac{\theta^\circ}{360^\circ} \times \pi r^2$ (where θ is in degrees) $= \frac{1}{2} r^2 \theta$ (where θ is in radians)
Segment		Area = $\frac{1}{2} r^2 (\theta - \sin \theta)$ (where θ is in radians)

Terms

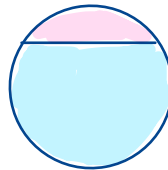
- Arc
 - major arc
 - minor arc



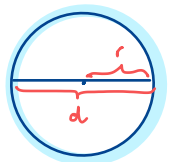
- sector
 - major sector
 - minor sector



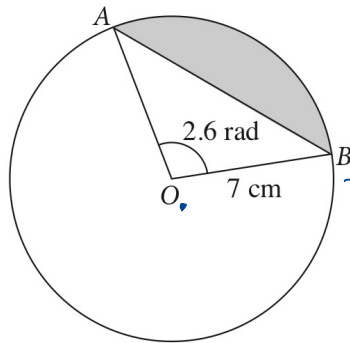
- segment
 - major segment
 - minor segment



- radius / radii
- diameter
- circumference



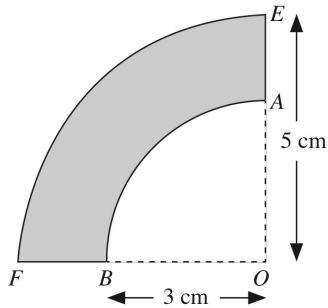
In the figure, O is the centre of the circle of radius 7 cm. AB is a chord and $\angle AOB = 2.6$ rad. The minor segment of the circle formed by the chord AB is shaded.



Find

- (a) the length of the minor arc AB ,
- (b) the area of the shaded region.

In the figure, the radii of quadrants ABO and EFO are 3 cm and 5 cm respectively.



- (a) Find the arc length of AB , in terms of π . $\approx$
- (b) Find the perimeter of the shaded region. Give your answer in the form $a + b\pi$.

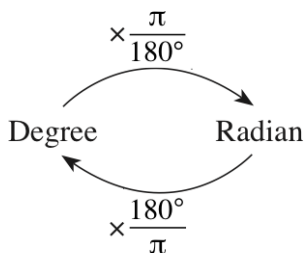
Degrees and Radians

2. π radians = 180°

$$1 \text{ radian} = \frac{180^\circ}{\pi}$$

$$1^\circ = \frac{\pi}{180^\circ} \text{ radians}$$

3. To convert from degrees to radians and from radians to degrees:



Conversion of Units

4. Length

$$1 \text{ m} = 100 \text{ cm}$$

$$1 \text{ cm} = 10 \text{ mm}$$

Area

$$\begin{aligned} 1 \text{ cm}^2 &= 10 \text{ mm} \times 10 \text{ mm} \\ &= 100 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} 1 \text{ m}^2 &= 100 \text{ cm} \times 100 \text{ cm} \\ &= 10\,000 \text{ cm}^2 \end{aligned}$$

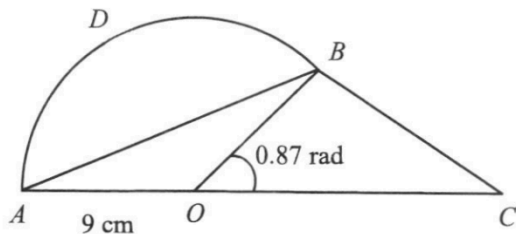
Volume

$$\begin{aligned} 1 \text{ cm}^3 &= 10 \text{ mm} \times 10 \text{ mm} \times 10 \text{ mm} \\ &= 1000 \text{ mm}^3 \end{aligned}$$

$$\begin{aligned} 1 \text{ m}^3 &= 100 \text{ cm} \times 100 \text{ cm} \times 100 \text{ cm} \\ &= 1\,000\,000 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} 1 \text{ litre} &= 1000 \text{ ml} \\ &= 1000 \text{ cm}^3 \end{aligned}$$

- Q 1. In the figure, $OADB$ is a sector of a circle with centre O and radius 9 cm. AOC is a straight line such that $\angle BOC = 0.87$ rad.



Calculate

- (a) the length of the arc ADB ,

Answer cm [2]

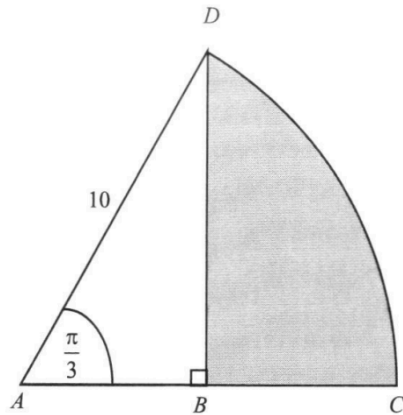
- (b) the area of segment ADB .

Answer cm^2 [3]

Q2.

The following diagram shows a sector ACD with radius 10 cm and

angle $DAC = \frac{\pi}{3}$ radian. B is a point on AC such that DB is perpendicular to AC .



- (a) Convert $\frac{\pi}{3}$ radians to degrees.

Answer [1]

- (b) Calculate the area of sector ACD .

Answer cm^2 [2]

- (c) Calculate the length AB .

Answer cm [2]

- (d) Find the perimeter of the shaded region BCD .

Answer cm [5]

Volume and Surface Area of Solids

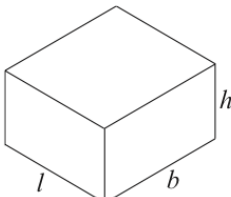
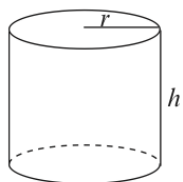
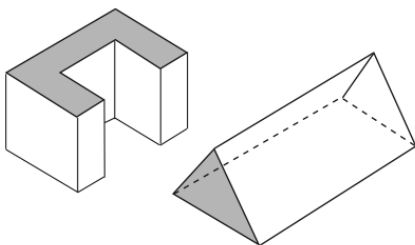
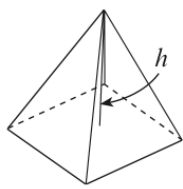
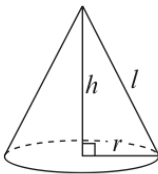
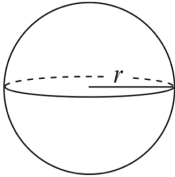
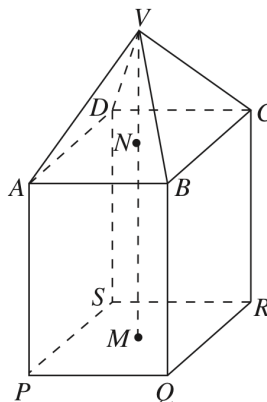
Figure	Diagram	Formulae
Cuboid		Volume = $l \times b \times h$ Total surface area = $2(lb + lh + bh)$
Cylinder		Volume = $\pi r^2 h$ Curved surface area = $2\pi r h$ Total surface area = $2\pi r h + 2\pi r^2$
Prism		Volume = Area of cross section $\times$ length Total surface area = Perimeter of the base $\times$ height + 2(base area)
Pyramid		Volume = $\frac{1}{3} \times \text{base area} \times h$
Cone		Volume = $\frac{1}{3} \pi r^2 h$ Curved surface area = $\pi r l$ (where l is the slant height) Total surface area = $\pi r l + \pi r^2$

Figure	Diagram	Formulae
Sphere		Volume = $\frac{4}{3}\pi r^3$ Surface area = $4\pi r^2$
Hemisphere		Volume = $\frac{2}{3}\pi r^3$ Surface area = $2\pi r^2 + \pi r^2$ = $3\pi r^2$



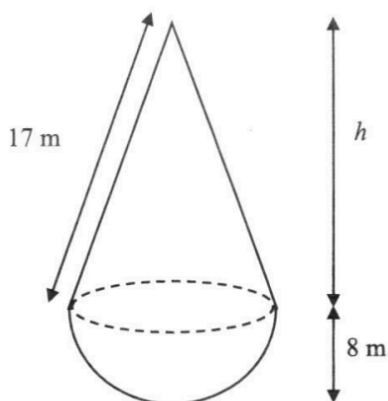
The diagram shows a solid which consists of a pyramid with a square base attached to a cuboid. The vertex V of the pyramid is vertically above M and N , the centres of the squares $PQRS$ and $ABCD$ respectively. $AB = 30$ cm, $AP = 40$ cm and $VN = 20$ cm.

- (a) Find
- (i) VA ,
 - (ii) $\widehat{VAC}$.
- (b) Calculate
- (i) the volume,
 - (ii) the total surface area of the solid.



Q2

The diagram below shows a solid made from a cone and a hemisphere.



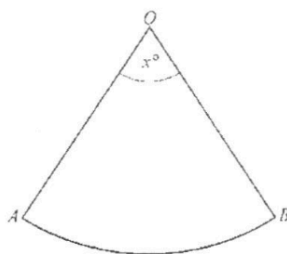
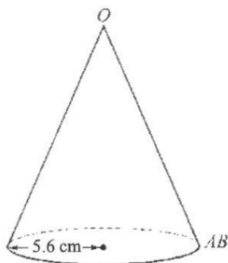
- (a) Show that the height, h , of the cone is 15 m.

[1]

- (b) Calculate the total surface area of the solid.

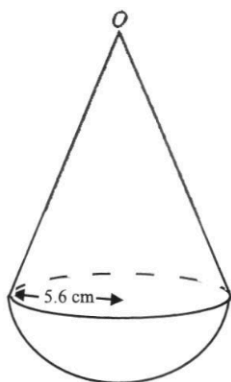
Answer m^2 [3]

- Q3. (a) The diagram below is a hollow cone of radius 5.6 cm and its volume is 259.44 cm^3 .
The cone is cut along the slant height from O to AB and is opened to form a sector OAB of a circle with centre O . Calculate the sector angle x° .



Answer $x = \dots\dots\dots$ [3]

- (b) Another cone in part (a) is joined to a solid hemisphere to form an ornament as shown below. Calculate the volume of the ornament.



Answer cm^3 [3]